SAI VENKAT

AI/ML & Data Platform Engineer | Lovely Professional University

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EDUCATION

Lovely Professional University - Phagwara, Punjab, India 2023 - 2027 [Expected]

B.Tech in Computer Science and Engineering (Specialization in AI & Data Engineering)

Cumulative GPA: 7.46 | Focus: Distributed Systems, ML Algorithms, LLM Reliability, Streaming Data

TECHNICAL COMPETENCIES

AI & LLM Systems: RAGAS, LangChain, HuggingFace, Local Vector Search, SentenceTransformers, Evaluation Harnesses

Distributed Systems: Redis Streams, FastAPI, Celery, Docker, PostgreSQL, Prometheus Telemetry, Task Queues

Data Engineering: Apache Kafka, dbt, Spark basics, Scikit-learn, SQL (Star Schema), Pandas, NumPy

Core Programming: Python, SQL, TypeScript / JavaScript, Bash, Git / GitHub Actions

SELECTED SYSTEMS PROJECTS

LLM Reliability & Evaluation Platform Python, RAGAS, LangChain, HuggingFace, FastAPI

- Deterministic evaluation harness benchmarking open-source LLMs under constrained compute using the RAG Triad.
- Formulated 3-stage validation pipeline calculating faithfulness, answer relevance, and context groundedness.
- Identified 28% variance in hallucination rates across zero-shot and few-shot reasoning prompts across 5 open models.

Distributed Asynchronous Task Queue Python, FastAPI, Redis Streams, Docker, PostgreSQL

- Asynchronous job execution platform featuring Redis Streams coordination, worker pools, and finite-state machines.
- Implemented exponential backoff with jitter and automated Dead-Letter Queue (DLQ) for poison-pill recovery.
- Demonstrated linear horizontal scaling across multiple worker instances with zero job loss during worker termination.

Distributed Workflow Orchestrator Python, FastAPI, AsyncIO, Docker, PostgreSQL

- Scalable DAG execution engine orchestrating multi-stage business pipelines across worker clusters.
- Built Kahn topological sorting algorithm for dynamic dependency resolution and transactional stage checkpointing.

Real-Time Streaming Fraud Detection Python, Kafka, Redis, Scikit-learn, FastAPI

- High-throughput transaction screening pipeline combining Kafka event streaming and Redis sliding-window caching.
- Built tri-state decision boundary (Approve / Challenge / Block) achieving sub-45ms p95 classification latency.

Business Analytics & Decision Intelligence Python, SQL, dbt, Pandas, Streamlit

- Analytical modeling platform with dbt dimensional models (star schema), SQL cohort aggregations, and RFM scoring.

ExoIntel: Exoplanet Discovery Platform Python, Lightkurve, Scikit-learn, NumPy

- ML pipeline on NASA Kepler/TESS transit photometry data for automated exoplanet candidate classification.

Curated portfolio, complete system architecture diagrams, and repositories verified at: https://github.com/saiiexd